Choosing Hamiltonians for `EqualNumberOfBitstrings.py`

a broad range of ground-state densities and initial overlaps

Recommendation in short

1. **Tilt the field** instead of choosing between B_z and B_x: draw an angle $\theta \sim U(0°, 90°)$ from the z axis and a strength $|B| = b\,|J|\max(1, \Delta)$ with b log-uniform in $[0.1, 10]$; set $B_x = |B|\sin\theta$, $B_z = |B|\cos\theta$. Keep $J \sim U(-1, 1)$.
2. **Use** $\Delta \in \{0,\ 0.5,\ 1,\ 2,\ 5,\ 10\}$. Drop $\Delta = 100$: almost every such cell is removed by the gap or trivial-state filter. Add 2 and 5, which give the intermediate densities.
3. **Select for density**, do not just sample: draw candidates from 1. and keep 2 per density bin of width 0.1, for every $(N, \text{dimension}, \Delta)$ (`select_hamiltonians.py`). This gives 20 Hamiltonians per group, spread evenly from density < 0.1 to > 0.9 at *every* N, and takes 30 s on a laptop.
4. **Initial overlaps:** with these Hamiltonians, the uniform draw from the support already spans 10^{-11}–10^{-2}. If large overlaps matter, draw the initial states stratified in $\log_{10}$ overlap. This raises the share above 10^{-2} from 6% to 15%.

1 Why the two current draws do not cover the range

The study's Hamiltonian is $H = J\sum_{\langle ij\rangle}(X_iX_j + Y_iY_j + \Delta Z_iZ_j) - \sum_i(B_xX_i + B_zZ_i)$ with $J, B \sim U(-1, 1)$ and $\Delta \in \{0, 0.5, 1, 10, 100\}$. The density counts basis states with $|\psi_i| > 10^{-3}$. Cells with gap $< 10^{-4}$ or top weight $\geq 1 - 10^{-10}$ are dropped. I drew 20 Hamiltonians per (N, dim, Δ) exactly as `enumerate_cells` does (Fig. 1, top two rows).

Longitudinal ($B_x = 0$). H conserves the total magnetisation, so the ground state lives in a single M_z sector. Its density can therefore never exceed $\binom{N}{N/2}/2^N$, which is 0.31 at $N = 6$ and 0.23 at $N = 12$. On top of that, 59% of the cells are dropped. For $J < 0$, $\Delta \geq 1$ the ferromagnet is a single basis state. For large Δ the two Néel states are split only at high order in $1/\Delta$.

Transversal ($B_z = 0$). The symmetry is broken and the densities move up: 40% of all cells are above 0.9. Low densities need an ordered state ($\Delta \gtrsim 10$), but there a transverse field splits the two ordered states only at high order (growing with N). So the gap closes and the filter removes them: 46% of the $\Delta = 10$ and 89% of the $\Delta = 100$ cells. Only 2% of the cells end up below 0.1. This is what the shards in `bitstring_shards_heisenberg_tv` show as well (201 of 424 kept cells above 0.9).

2 What sets the density

Figure 2 maps the density at $N = 12$ over the two field parameters, from a scan of 11 520 Hamiltonians ($N = 6$–12, 1D and 2D, nine Δ, both signs of J, ten field strengths, eight angles; `scan_grid.py`).

- **Strong field ($b \gg 1$):** the ground state approaches the product state along the field. Its amplitudes are $\cos^{N-k}(\theta/2)\sin^k(\theta/2)$, so the angle alone sets the density: at $N = 12$ it is 0.02, 0.07, 0.19, 0.61, 0.98 for $\theta = 5°, 15°, 30°, 45°, 60°$. The gap is of order $|B|$, so all of these cells survive. This is the clean knob for the whole range.
- **Weak field, $J > 0$, $\Delta > 1$:** Néel-like ground states with densities of a few per cent that survive the gap filter up to $\Delta \approx 10$. These give the low end without being product states. At $\Delta = 100$ they are lost (the gap is below 10^{-4}).
- **Weak field, $J < 0$:** for $\Delta \geq 1$ the ferromagnetic doublet is degenerate (pure B_x) or a single basis state (pure B_z). Any tilt between the two lifts both problems, which is why the tilted draw keeps 98–99% of the cells.
- The field has to compete with the largest coupling, $|J|\max(1, \Delta)$, and not with $|J|$ alone. Otherwise the high densities are out of reach at $\Delta = 10$ on the 2D lattices. Hence the factor $\max(1, \Delta)$ in the recommended field strength.
- At $\Delta = 1$ the model is SU(2) symmetric. The field angle only rotates the basis, so for weak fields the density does not depend on θ (the singlet is rotation invariant).

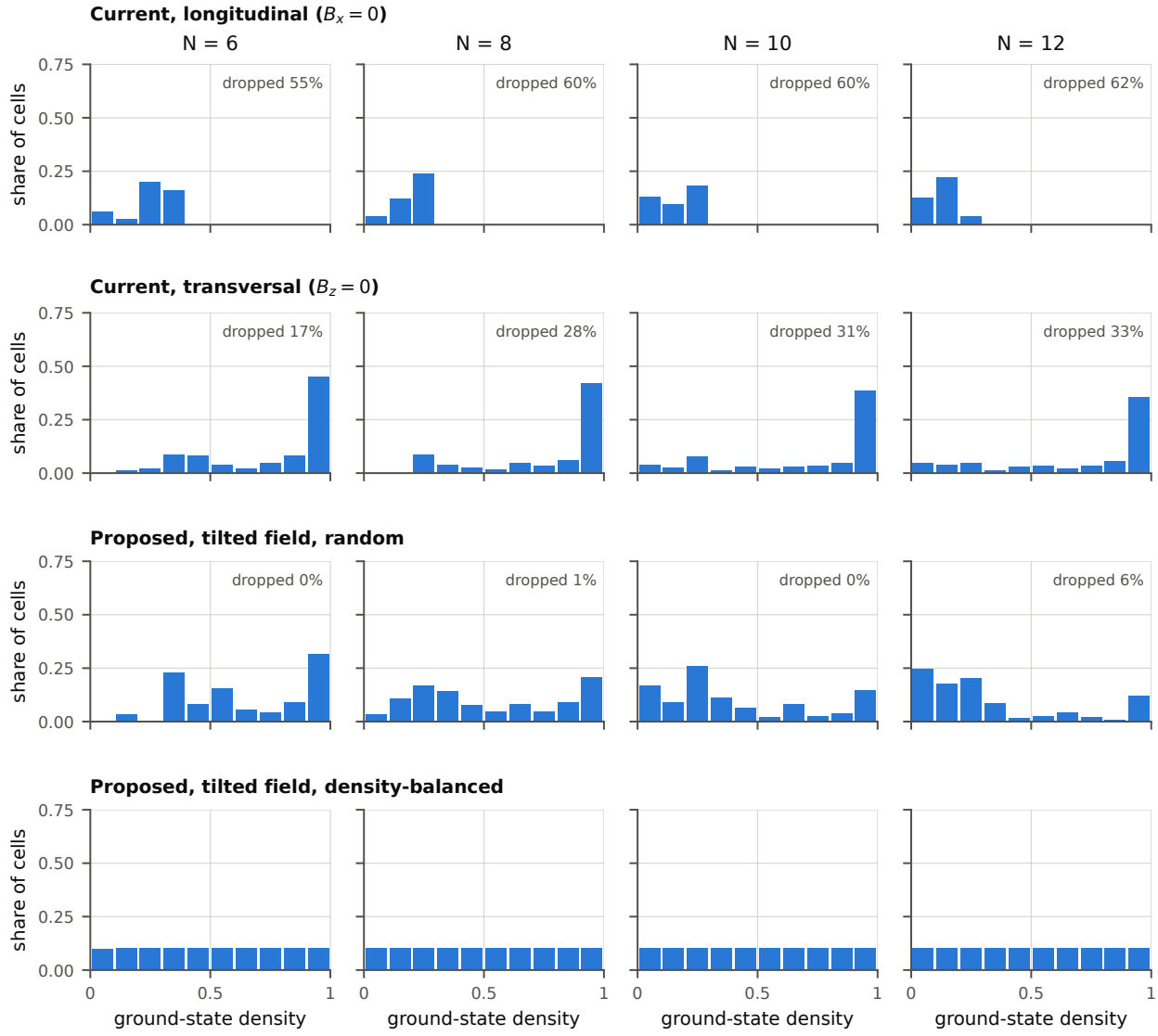


Figure 1: Ground-state densities of the cells that survive the gap and trivial filters. Bars are shares of all drawn cells, so a row sums to the kept fraction. Each row is one way of drawing the Hamiltonians: 20 per $(N, \dim, \Delta)$, both geometries pooled. Only the density-balanced selection is flat at every N.

3 Recommended parameters

Random tilted field. This is the minimal change to `enumerate_cells`. Keep J_1 the first draw, so that the longer-range couplings stay where they are:

```
J_1, log_b, theta = rng.uniform(-1, 1), rng.uniform(-1, 1), rng.uniform(0, np.pi / 2)
...
B = 10 ** log_b * abs(J_1) * max(1.0, delta)     # per cell, since it depends on delta
Bx, Bz = np.round(B * np.sin(theta), 6), np.round(B * np.cos(theta), 6)
```

This keeps 98% of the cells and spans < 0.1 to > 0.9 for $N \geq 8$, but the distribution is still uneven (Fig. 1, third row). At $N = 6$ no cell falls below 0.1, and at $N = 12$ few fall between 0.4 and 0.9. That is a consequence of the fixed threshold $|\psi_i| > 10^{-3}$: a flat state has amplitude $2^{-N/2}$, so the same ground state counts differently at different N.

Density-balanced selection (recommended). `select_hamiltonians.py` draws from the scheme above and keeps the first two Hamiltonians that land in each density bin $[0, 0.1), \ldots, [0.9, 1]$ and pass the study's filters. It does this for each $N \in \{6, 8, 10, 12\}$, both geometries and each Δ. It filled 959 of the 960 slots from 32 777 candidates in 32 s. The one gap is the lowest bin at $N = 6$, 2D, $\Delta = 0$, where a density below 0.1 means at most six of 64 states. The resulting table (`data/selected_hamiltonians.csv`) replaces the draw in `enumerate_cells`:

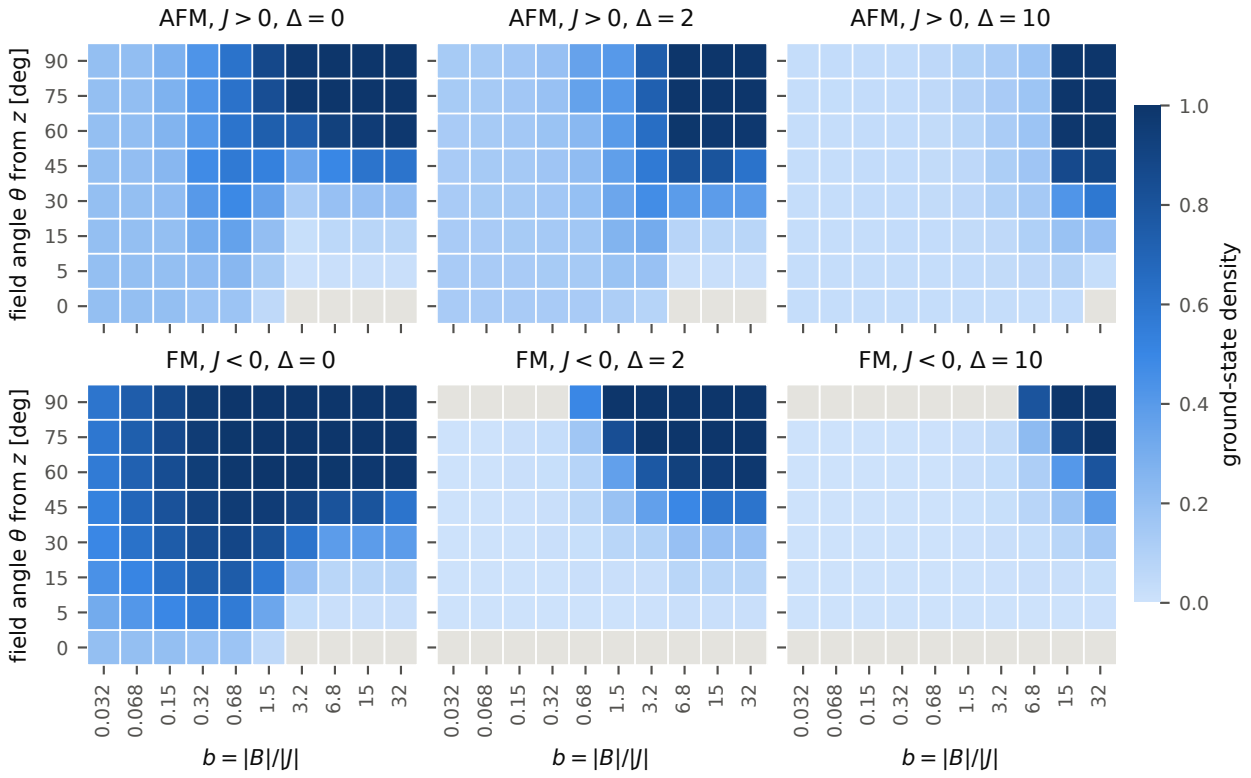


Figure 2: Ground-state density at $N = 12$ (1D) over field strength $b = |B|/|J|$ and angle θ from the z axis ($0°$: longitudinal, $90°$: transversal). Grey cells are removed by the study's filters. The current study only uses the bottom row ($\theta = 0$) or the top row ($\theta = 90°$) of each panel.

```
table = pd.read_csv("StudyOfHamiltonians/data/selected_hamiltonians.csv")
return [Cell(int(r.hamiltonian_index), int(r.num_sites), int(r.dimension), float(r.delta),
             (float(r.J),), float(r.Bx), float(r.By), float(r.Bz), 1)
        for r in table.itertuples()]
```

Trade-off: the Hamiltonian index no longer refers to the same couplings across N and Δ, so paired comparisons across N are lost. Comparisons within a cell are unchanged: BARK and SKQD still run on the same Hamiltonian and initial states.

Initial overlaps. Overlap and density cannot be chosen independently. A flat ground state has every overlap near 2^{-N}, while a concentrated one offers overlaps from 10^{-12} up to ≈ 1 (Fig. 3, bottom). The broad density range is therefore what broadens the overlaps. With the balanced selection, the current uniform draw from the support covers $10^{-10.7}$–$10^{-1.8}$ (5–95%), against $10^{-7.9}$–$10^{-0.8}$ for the current longitudinal study. If a wide range *within* each Hamiltonian is the goal, draw one state per equal-width bin of $\log_{10} p$ between the smallest and largest overlap of the support (`stratified` in `simulate_schemes.py`). This covers $10^{-10.6}$–$10^{-0.5}$, with 15% instead of 6% above 10^{-2}. The averages over initial states are then over a designed spread rather than a typical starting point. They should be read per overlap bin, which the `overlap` facet already does.

Smaller points.

- The field is now two numbers. Add a facet for $\theta = \arctan(B_x/B_z)$ or for B_x next to `Bz` in `FACETS`, otherwise the `Bz` folders mix very different fields.
- 11% of the selected Hamiltonians are near-product states (top weight > 0.99), and the smallest gap is 3×10^{-4}, just above the cut. If either worries you, tighten `MIN_GAP` or the trivial cut in `select_hamiltonians.py` as well, so that the table and the plots agree.
- The study here covers `max_distance` $= 1$ and $B_y = 0$, as in the current runs.

Reproducing. In `StudyOfHamiltonians/`: `scan_grid.py` (grid, Fig. 2), `simulate_schemes.py` (current and candidate draws), `select_hamiltonians.py` (balanced table), `make_figures.py`, then `pdflatex report.tex`. Nothing outside the folder is modified. `RandomSpinModel.py` is only imported.

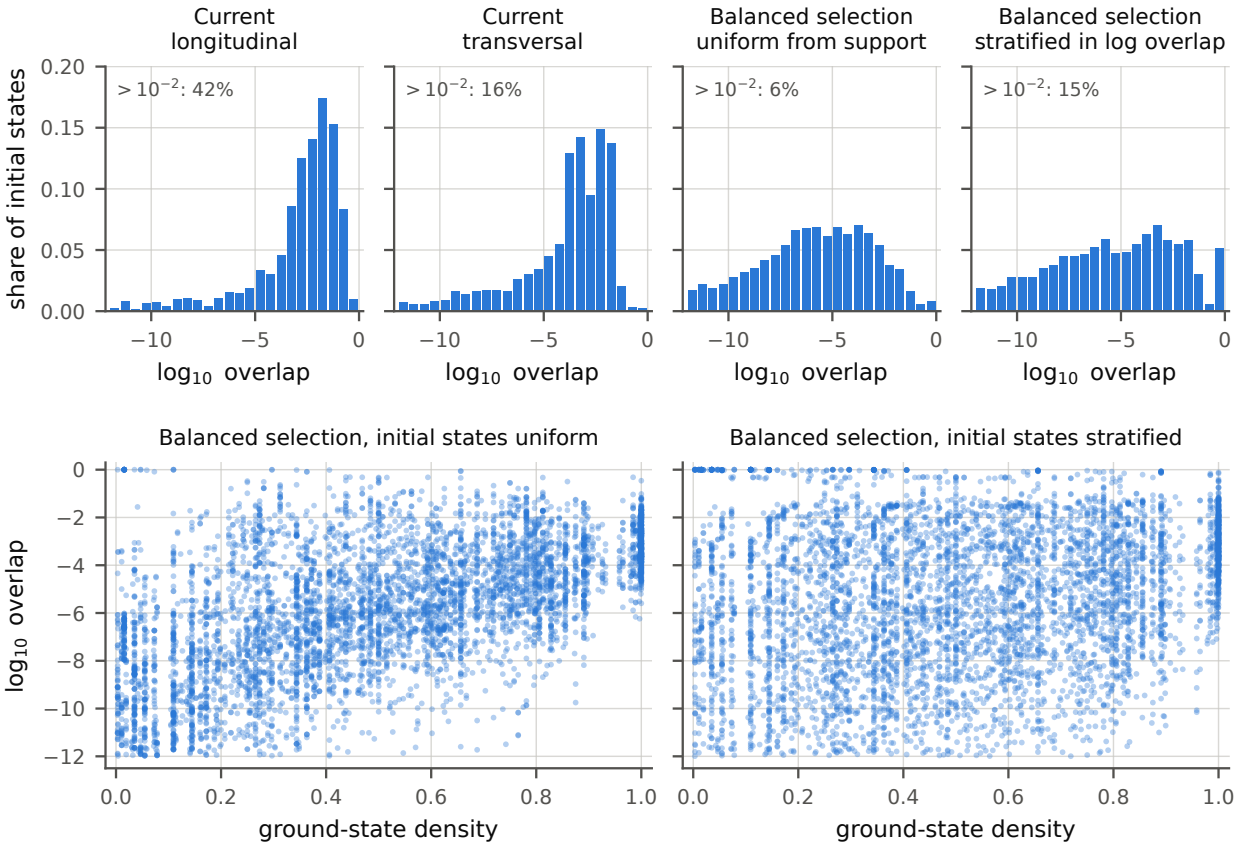


Figure 3: Top: overlaps of the initial states, six per kept Hamiltonian. Bottom: overlap against ground-state density for the balanced selection. Each column of points is one Hamiltonian. Dense ground states can only offer overlaps near 2^{-N}.